

Pre-Board Examination - 2079

Class : XII

Subject: Mathematics (0081)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group 'A'

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

- $\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots$ to ∞ is
 a) e b) $2e$ c) $e - 1$ d) $\frac{2n}{n!}$
- The system of equations $x + 2y = 5$ and $10 - 2x - 4y = 0$ have
 a) no solution b) unique solution
 c) two solutions d) infinitely many solutions
- $\sin^{-1}x$ is equal to
 a) $\frac{1}{\sin x}$ b) $\frac{\pi}{2} - \cos^{-1}x$ c) $\sqrt{1 - (\cos^{-1}x)^2}$ d) $\frac{\pi}{2} + \cos^{-1}x$
- If $\cos p\theta = \cos q\theta$, $p \neq q$, then θ is
 a) $2n\pi$ b) $\frac{2n\pi}{p-q}$ c) $\frac{2n\pi}{p+q}$ d) $\frac{2n\pi}{p \pm q}$
- If $\vec{a}$ and $\vec{b}$ are diagonals of a parallelogram, then its area is
 a) $\frac{1}{2}|\vec{a}| \times |\vec{b}|$ b) $\frac{1}{2}|\vec{a} \times \vec{b}|$ c) $|\vec{a} \times \vec{b}|$ d) $\frac{1}{2}|\vec{a} \times \vec{b}| \sin\theta$
- In a binomial distribution $(q + p)^n$, the probability of occurrence of r successes out of n trials is
 a) ${}^nC_r p^{n-r} q^r$ b) ${}^nP_r q^{n-r} p^r$ c) ${}^nC_{n-r} q^{n-r} p^r$ d) ${}^rC_n q^{n-r} p^r$
- The distance between centre and vertex of an ellipse is
 a) radius b) semi-major axis c) major axis d) minor axis

8. $\int \frac{e^x dx}{1+e^{2x}}$ is equal to

- a) $\tan^{-1}(e^x) + C$ b) $\ln(1 + e^{2x}) + C$
c) $\tan^{-1}(e^{2x}) + C$ d) $\ln \frac{1+e^x}{1-e^x} + C$

9. The degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is

- a) 1 b) 2 c) 3 d) 4

10. For the linear programming problem, we use the simplex method to find the solution which is

- a) feasible or optimal b) feasible but not optimal
c) optimal but not feasible d) feasible and optimal

11. P and Q are two unlike parallel forces such that Q is one third of P. What is their resultant?

- a) 2Q b) 2P c) $\frac{3P-Q}{3}$ d) $\frac{P-3Q}{3}$

OR

The order of the difference equation $y_t + 2y_{t-1} - y_{t-2} = 0$ is

- a) 0 b) 1 c) 2 d) 3

Group 'B'

Short Answer Questions.

[8 × 5 = 40]

12. a) In how many ways can 3 different books of Physics and 4 different books of Mathematics be kept in a book shelf? In how many ways can it be done if all books of same subjects are to keep together? [1+2]
b) Using principle of mathematical induction, prove that:
 $2 + 4 + 6 + \dots + 2n = n(n + 1)$ [2]
13. a) If the roots of the equation $ax^2 + bx + c = 0$ be in the ratio 3:4, prove that: $12b^2 = 49ac$. [2]
b) Let $G = \{0, 1, 2\}$. Form a composition table for G under addition modulo 3. Find the inverse of each elements. [3]

14. a) Find the eccentricity and the coordinates of foci of the hyperbola $x^2 - 4y^2 - 12 = 0$. Also, obtain the distance between two foci. [2+1]
 b) Find a unit vector perpendicular to the plane determined by the vectors $\vec{a} = \vec{i} + \vec{j} - 2\vec{k}$ and $\vec{b} = \vec{i} - 2\vec{j} + \vec{k}$. [2]
15. a) The regression coefficients of x on y and y on x are 0.84 and 0.32 respectively. If the arithmetic means of x and y are 42 and 26 respectively. Find the two equations of lines of regression and the correlation coefficient between x and y. [3]
 b) Suppose that 5 students from Physical group and 6 students from Biological group are interested to participate in a group dance of 5 for the annual farewell programme. What is the probability that the group contains 2 students from Physical group and 3 from Biological group? [2]
16. a) Find from first principle, the derivative of $\ln(\cos x)$. [3]
 b) Evaluate : $\int \frac{dx}{x(1-x)}$ [2]
17. a) Evaluate : $\int \frac{dx}{1-2\cos x}$ [3]
 b) Using L'Hospital's rule, evaluate: $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec 3x}{\sec x}$ [2]
18. a) Using Gauss-Elimination Method, solve the equations $x + 2y = 5$, $5x - 3y = -1$ [2]
 b) Using simplex method, maximize $Z = 6x - 9y$ subject to the constraints: $2x - 3y \leq 6$, $x + y \leq 20$, $x, y \geq 0$ [3]
19. Find the velocity and direction of projection of a shot which passes in a horizontal direction just over the top of a wall which is 250 m off and 125 m high. ($g = 9.8 \text{ ms}^{-2}$) [5]

OR

The demand function for a good is $P = 60 - 2Q$, fixed cost is Rs.192 and the variable cost for each additional unit is Rs.20.

- a) Write down the equations for total revenue and total cost in terms of Q.
 b) Find break-even point.
 c) Find the TR and TC at the break-even points. [2+1+2]

Group 'C'

Long Answer Questions.

[3 × 8 = 24]

20. a) Show that the middle term in the expansion of $(1 + x)^{2n}$ is $\frac{1.3.5 \cdots (2n-1)}{n!} 2^n \cdot x^n$ [3]
- b) If $z = \cos \theta + i \sin \theta$, prove that:
i) $z^n + \frac{1}{z^n} = 2 \cos n\theta$ ii) $\text{Arg}(\bar{z}) = 2\pi - \text{Arg}(z)$ [2+1]
- c) Find the n^{th} term and sum of the first 10 terms of the series
 $1.1 + 2.4 + 3.9 + 4.16 + \cdots$ [2]
21. a) Solve for general values of x : $\cos x + \cos 2x + \cos 3x = 0$ [3]
- b) Mention the condition for the two planes to be parallel to each other. Find the equation of plane through the intersection of the planes
 $x + y + z = 6$ and $2x + 3y + 4z + 5 = 0$ and perpendicular to the plane $4x + 5y - 3z = 8$. [1+4]
22. a) Examine whether the function $f(x) = x^2 - 2x$ satisfies Lagrange's Mean Value Theorem [MVT] on $[1,4]$. If it satisfies MVT, find the coordinates of the point at which the tangent is parallel to the chord joining the points $A(1, -1)$ and $B(4, 8)$. [3+1]
- b) Identify whether each of the given differential equation is linear or homogeneous? Solve any one of them:
i) $x dy - y dx = \sqrt{x^2 + y^2} dx$ ii) $\cos^2 x \frac{dy}{dx} + y = 1$ [1+3]